

Q1 - 24 January - Shift 1

A spherical body of mass 2 kg starting from rest acquires a kinetic energy of 10000 J at the end of 5th second. The force acted on the body is ____ N.

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Q2 - 24 January - Shift 2

A body of mass 1 kg begins to move under the action of a time dependent force $\vec{F} = (t\hat{i} + 3t^2\hat{j})\text{N}$,

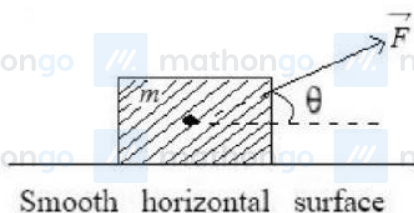
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where $\hat{i}$ and $\hat{j}$ are the unit vectors along x and y axis. The power developed by above force, at the time $t = 2\text{s}$, will be ____ W.

Q3 - 25 January - Shift 1

An object of mass 'm' initially at rest on a smooth horizontal plane starts moving under the action of force $F = 2\text{N}$. In the process of its linear motion, the angle θ (as shown in figure) between the direction of force and horizontal varies as $\theta = kx$, where k is a constant and x is the distance covered by the object from its initial position. The expression of kinetic energy of the object will be $E = \frac{n}{k} \sin \theta$. The value of n is ____.

Space for your notes:



Q4 - 29 January - Shift 1

A 0.4 kg mass takes 8s to reach ground when dropped from a certain height 'P' above surface of earth. The loss of potential energy in the last second of fall is ____ J. [Take $g = 10 \text{ m/s}^2$]

Space for your notes:

Q5 - 29 January - Shift 2

Identify the correct statements from the following:

(A) Work done by a man in lifting a bucket out of a well by means of a rope tied to the bucket is negative.

(B) Work done by gravitational force in lifting a bucket out of a well by a rope tied to the bucket is negative.

(C) Work done by friction on a body sliding down an inclined plane is positive.

(D) Work done by an applied force on a body moving on a rough horizontal plane with uniform velocity is zero.

(E) Work done by the air resistance on an oscillating pendulum is negative.

Choose the correct answer from the options given below:

(1) B and E only (2) A and C only

(3) B, D and E only (4) B and D only

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Q6 - 30 January - Shift 2

A body of mass 2 kg is initially at rest. It starts moving unidirectionally under the influence of a source of constant power P. Its displacement in 4s is $\frac{1}{3} \alpha^2 \sqrt{P} \text{ m}$. The value of α will be

Space for your notes:

Q7 - 01 February - Shift 1

A small particle moves to position $5\hat{i} - 2\hat{j} + \hat{k}$ from its initial position $2\hat{i} + 3\hat{j} - 4\hat{k}$ under the action of force $5\hat{i} + 2\hat{j} + 7\hat{k}$ N. The value of work done will be _____ J.

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Q8 - 01 February - Shift 2

A force $F = (5 + 3y^2)$ acts on a particle in the y-direction, where F is newton and y is in meter. The work done by the force during a displacement from $y = 2\text{m}$ to $y = 5\text{m}$ is _____ J.

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Q9 - 01 February - Shift 2

A block is fastened to a horizontal spring. The block is pulled to a distance $x = 10\text{ cm}$ from its equilibrium position (at $x = 0$) on a frictionless surface from rest. The energy of the block at $x = 5\text{ cm}$ is 0.25 J . The spring constant of the spring is _____ Nm^{-1} .

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Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (40)**Q2** (100)**Q3** (2)**Q4** (300)**Q5** (1)**Q6** (4)**Q7** (40)**Q8** (132)**Q9** (50)

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Q1 (40)

$$\frac{1}{2} \times 2 \times v^2 = 10000$$

$$\Rightarrow v^2 = 10000$$

$$\Rightarrow v = 100 \text{ m/s}$$

$$\Rightarrow v = at = a \times 5 = 100$$

$$\Rightarrow a = 20 \text{ m/s}^2$$

$$F = ma = 2 \times 20 = 40 \text{ N}$$

Q2 (100)

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$$\vec{F} = t\hat{i} + 3t^2\hat{j}$$

$$\frac{m d\vec{v}}{dt} = t\hat{i} + 3t^2\hat{j}$$

$$m = 1\text{kg}, \int_0^t dv = \int_0^t t dt \hat{i} + \int_0^t 3t^2 dt \hat{j}$$

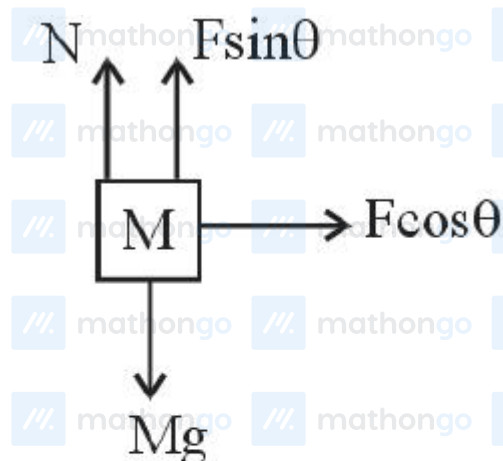
$$\vec{v} = \frac{t^2}{2}\hat{i} + t^3\hat{j}$$

$$\text{Power} = \vec{F} \cdot \vec{v} = \frac{t^3}{2} + 3t^5$$

$$\text{At } t = 2, \text{ power} = \frac{8}{2} + 3 \times 32$$

$$= 100$$

Q3 (2)



$$F \cos \theta = ma$$

$$2 \cos(kx) = \frac{mvdv}{dx}$$

$$\int_0^v v dv = 2 \int_0^x \cos(kx) dx$$

$$\frac{mv^2}{2} = \frac{2}{k} \sin kx$$

$$\text{K.E.} = \frac{2}{k} \sin \theta$$

$$n = 2$$

Q4 (300)

Displacement is 8^{th} sec.

$$S_8 = 0 + \frac{1}{2} \times 10 \times (2 \times 8 - 1)$$

$$S_8 = 5 \times 15$$

$$\Delta U = 0.4 \times 10 \times 5 \times 15$$

$$\Delta U = 20 \times 15 = 300$$

Q5 (1)

No Solution Available

Q6 (4)

$$\frac{1}{2} m V^2 = P t$$

$$V = \sqrt{\frac{2 P t}{m}}$$

$$\frac{dx}{dt} = \sqrt{\frac{2 P t}{m}}$$

$$x = \sqrt{\frac{2 P}{m}} \frac{2}{3} [t^{3/2}]_0^4$$

$$x = \frac{16 \sqrt{P}}{3} = \frac{1}{3} \times 16 \sqrt{P}$$

$$\alpha = 4$$

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Q7 (40)

$$W = \vec{F} \cdot (\vec{r}_f - \vec{r}_i) \\ = (5\hat{i} + 2\hat{j} + 7\hat{k}) \cdot ((5\hat{i} - 2\hat{j} + \hat{k}) - (2\hat{i} + 3\hat{j} - 4\hat{k}))$$

$$W = 40 \text{ J}$$

Q8 (132)

$$F = 5 + 3y^2$$

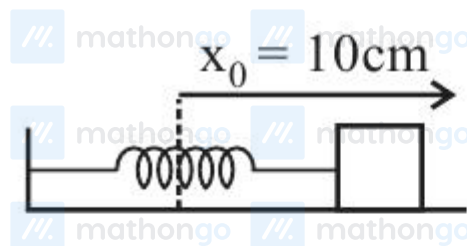
$$W = \int_2^5 (5 + 3y^2) dy$$

$$= \left[5y + \frac{3y^3}{3} \right]_2^5$$

$$= 132 \text{ J}$$

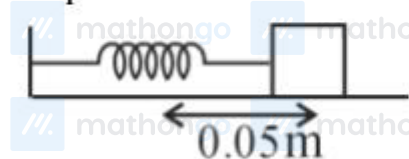
Q9 (50)

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$$U_i = \frac{1}{2} k x_0^2$$

$$K_i = 0$$



$$U_f = \frac{1}{2} k \left(\frac{x_0}{2} \right)^2$$

$$K_f = 0.25 \text{ J}$$

$$\frac{1}{2} k x_0^2 + 0 = \frac{1}{2} k \frac{x_0^2}{4} + 0.25$$

$$\frac{1}{2} k x_0^2 \frac{3}{4} = \frac{1}{4}$$

$$\frac{1}{2} k \frac{3}{100} = 1 \Rightarrow k = \frac{200}{3} \text{ N/m}$$

$$= 67 \text{ N/m}$$